EXERCISE SET 12.4 — SOLUTIONS

Ganita Manjari | Grade 9 | Chapter: Quadrilaterals (Tiling the Plane)

READING NOTE: A CUTTING-EDGE DISCOVERY

The shape known as the **"hat"**—discovered in 2023 by a team of four mathematicians (Smith, Myers, Kaplan, and Goodman-Strauss)—is the first single shape that leads to *aperiodic tilings*. When identical copies of the hat tile the plane endlessly, they do so without ever repeating the exact same pattern.

Question 1

Justify why the plane cannot be tiled with a regular pentagon.
(Hint: Read the first 3 sentences of 'Think and Reflect' in the section on tiling.)
(Note: While regular pentagons cannot tile the plane, certain irregular pentagons can. The 15th and most recent type was discovered in 2015.)

SOLUTION & EXPLANATION:

1. **Mathematical Condition for Tiling around a Vertex:**
   To tile a 2D plane completely without gaps or overlaps, the interior angles of the shapes meeting at any vertex must add up to exactly **360°**.
2. **Interior Angle of a Regular Pentagon:**
   The formula for the interior angle of a regular n-gon is:

   $$\text{Angle} = ((n - 2) \times 180°) / n$$

   For a regular pentagon ($n = 5$):

   $$\text{Angle} = ((5 - 2) \times 180°) / 5 = (3 \times 180°) / 5 = 108°$$

3. **Checking Vertex Angle Sum:**
   If we fit regular pentagons together at a common vertex:
   - 3 pentagons at a vertex: $3 \times 108° = 324°$ (Leaves a gap of $36°$).
   - 4 pentagons at a vertex: $4 \times 108° = 432°$ (Overlaps by $72°$).
4. **Conclusion:**
   Since 360° is **not divisible** by 108° (i.e., $360° / 108° = 3.33...$ is not an integer), regular pentagons cannot fit together seamlessly around a vertex. Therefore, the plane **cannot** be tiled using regular pentagons.

Question 2

Draw a non-convex 4-gon DART. Show how we can tile the plane with copies of DART. Both methods that we discussed earlier will work. Which do you prefer?

SOLUTION & CONSTRUCTION GUIDE:

1. Understanding a Non-Convex 4-gon (Dart):

- A non-convex 4-gon (quadrilateral) has one interior angle greater than 180° (a reflex angle). It resembles an arrowhead or a "dart".
- The sum of all 4 interior angles of any quadrilateral is always **360°**.

2. Two Methods to Tile the Plane with DART copies:

> **Method 1: Translation & Rotation (Pairing into Parallelograms/Hexagons)**
> Join two identical DART shapes along their matching sides to form a convex polygon (like a parallelogram or a symmetric hexagon). Since any parallelogram can tile the plane easily by shifting (translating) horizontally and vertically, copies of the DART pair will cover the entire plane seamlessly.

> **Method 2: Vertex-Fitting Rule (4-Color/Angle Rotation)**
> Place 4 identical copies of the DART around a single vertex such that each of the 4 different angles of the quadrilateral meets at that point. Since the four interior angles of any quadrilateral sum to 360°, they will fit together perfectly around every vertex without gaps.

3. Preference & Comparison:

- **Preference:** Method 1 (Pairing into Parallelograms) is generally preferred for practical drawing because shifting identical composite blocks in straight rows and columns makes grid-building very straightforward and visually neat.